

The Value PCs

Rs 25,000 is not just a nice, round figure—it's also a good cut-off point to segregate PCs, going by current prices. This segment is primarily targeted at beginning home users, work-at-home individuals, students, etc.

We should emphasise here that based on what we've seen in this test, a sub-25K PC is not to be dismissed as a "cheap PC"—several models that fell in this category offer you very decent features indeed!

In this segment are featured seven PCs: the Compaq Presario SR1502IL, Compaq Presario SR1008IL, Datamini Festiva MG Value, HCL Ezeebee PI 873, HCL Ezeebee 4981, Zenith Premium 7D, and the LG XT1-VH1SHP1.

Features

Of the seven PCs mentioned above, two were AMD-based, and five were Intel-based. It was good to see a branded PC in this segment—the HCL EzeeBee PI 873—featuring the powerful AMD Athlon64 processor. But this wasn't the only PC in this segment with a powerful processor: the Zenith Premium 7D sported a 3.0 GHz Intel Pentium 4 with HyperThreading Technology, which is something we didn't expect in this, the value category. These are high-end processors, and

these two PCs therefore bring powerful number-crunching capabilities at an affordable price.

We also saw some of the best hard drives being featured on some of these PCs. While most sported the older and slower IDE interface, the HCL PCs came equipped with the faster SATA drives. In terms of drive capacities too, HCL lead the pack with 74 GB and 80 GB drives, whereas the others stopped at 40 GB—which these days is considered the bare minimum (read "almost inadequate").



Compaq Presario SG1008IL

In the memory department, once again, only the HCL PCs came with as much as 512 MB of RAM. The Compaq, LG and Zenith PCs came with just 128 MB of RAM, and since this was shared by the onboard graphics, heavy performance penalties were obvious: we could actually see these PCs crawl-

ing compared to others with larger memories.

The HCL and Zenith PCs came with Combo drives, while the rest came with plain CD-ROM drives. We should mention that the Datamini PC came with the sloppiest CD-ROM drive we've ever seen—it had a hard time reading even its own driver CD, and we had to connect our own optical drive to install the OS, drivers and benchmark applications! The Compaq SR1502IL and the Zenith PCs had optical drives more or less similar to that of the Datamini.

As we'd expected, none of the PCs in this segment featured a separate graphics card, going for onboard solutions instead. Of these, the Intel GMA 900 is the most feature-rich, and is DirectX 9 compliant—which means that it is ready, out-of-the-box, for the next version of Windows—Vista. It is also good for moderate gaming, and is markedly faster than the other onboard solutions in this category such as the SiS 330 Mirage and Intel Extreme Graphics. Of the seven PCs in this category, the HCL Ezeebee 4981, Zenith Premium 7D, and the Datamini Festiva MG Value featured Intel GMA 900.

Micro ATX cabinets were the choice of most brands, while the Datamini provided an ATX cabinet—albeit of an inferior build

How We Tested

As in all our tests, the home PCs reviewed here were rigorously benchmarked and tested. We chose the benchmarks and tests considering they needed to be run on a range of computers, from the least powerful to the most powerful. Windows XP Pro with SP1 was freshly loaded, along with DirectX 9.0c, and the drivers from the driver CDs provided with the PC (or from the manufacturer's Web site, if a driver CD was not provided) were loaded. This was done even on the Media Centre PCs, because the Windows XP Media Center 2005 OS that ships with these PCs is not available off-the-shelf—and the version that came on the CDs provided comes with SP2, on which one of our benchmarks doesn't run.

The tests and benchmarks consisted of the following.

FarCry: This is a Direct3D game that accurately measures the Direct3D gaming subsystem. We chose *FarCry* because it uses a fairly modern graphics engine and reflects how future games will play on the system.

Call of Duty: This OpenGL game is used to test the strength of the OpenGL graphics subsystem. We chose *Call of Duty* because it runs easily even on motherboards with onboard graphics.

PC Mark 2004 (PCMark04): This is an industry-standard benchmark that measures the CPU, memory, graphics and hard disk performance of the system, and hence is a good system-wide benchmark.

SiSoft Sandra 2005 SR3: This consists of benchmark modules that measure CPU, filesystem and memory performance.

Ziff-Davis Business Winstone 2004: This benchmark suite runs various applications such as WinZip, Excel 2002, PowerPoint 2002, etc., to measure the real-world performance of a system. It returns a score that is comparable with any system that runs this test suite.

Dr. DivX 1.06: This is a video encoding application used to encode DivX files. We encoded a 100 MB MPEG2 file to DivX and noted the time taken to encode.

DBPowerAmp: This is an audio conversion tool used to encode audio files. We converted a 200 MB WAV file to a VBR MP3 file at the highest possible quality, and noted the time taken to encode.

DisplayMate: We used DisplayMate Video Edition to evaluate the quality of the monitor by gauging the sharpness and clarity of the display.

Nero CD-DVD Speed: We used this application bundled with Nero Burning ROM to create a 700 MB test data CD, and then used the CD to test the average speed of the optical drive and also its access time.

Speaker Test: Various types of audio files were played on the speakers that came with the PCs (if any did), and we noted the quality of the audio.

In addition to these tests and benchmarks, features such as the number and type of optical and hard drives, monitor type and size, mouse and keyboard type, ports, cabinet type, front connector options such as USB and card readers, airflow considerations, and bundled accessories such as speakers and gamepads were taken into account before we decided on each PC's overall score.